VISHWAKARMA AWARDS
A Maker Bhavan Foundation Initiative

MAKER BHAVAN FOUNDATION

VISHWAKARMA AWARDS 2026-27

# Vishwakarma Application ID

## VK26-AU-FD-0015

**NadiSense**: a ring that checks your pulse, oxygen and temperature while you sleep

Frontier Diagnostics: The Next Vital Signs, Team Ring of Care, Rajalakshmi Engineering College

# Problem Context & Project Overview

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| 1 in 4 | 2 of 3 | >85% | ~80% |
|---|---|---|---|
| ADULTS HYPERTENSIVE | OF THEM UNDIAGNOSED | SLEEP APNEA UNDIAGNOSED | RURAL SPECIALIST SHORTFALL |

*Sources: NFHS-5 analyses (medRxiv 2024); ResMed economic model, ISPOR Europe 2025; MoHFW Health Dynamics of India 2022-23.*

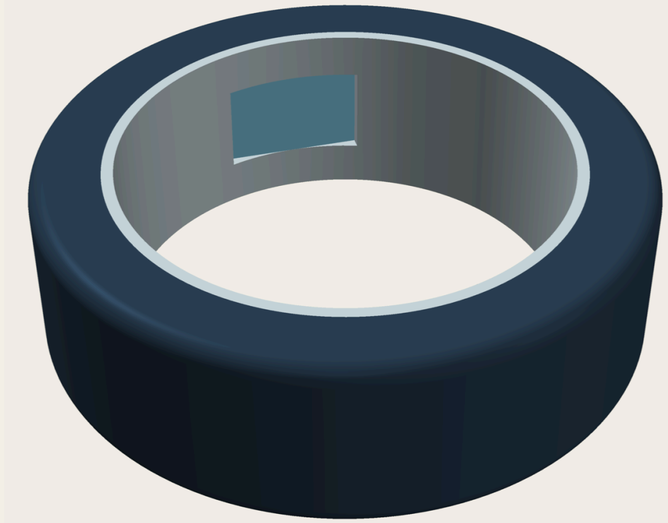


*Figure: NadiSense ring v2, CAD render (AI Generated)*

## Target Beneficiary & Problem Statement

- **Target User:** adults aged 30–60 in semi-urban and rural India, starting with Chennai's peri-urban PHC/CHC areas.
- **Pain Point:** hypertension, irregular heart rhythm and sleep apnea are only caught by one-off tests: a BP cuff, an ECG or an overnight sleep study.
- **Gap:** nobody measures at night, and rural clinics often have no specialist to run these tests.
- **Existing Solutions:** current smart rings are fitness gadgets for city users, with no link to a PHC referral.

**Project:** NadiSense | Track: Frontier Diagnostics | Core idea: a pulse, $SpO_2$ and temperature ring worn overnight.

# User Journey: Before State

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| 1 in 11 | < 1 in 5 | ₹19,500 Cr | 4,413 / 21,964 |
|---|---|---|---|
| HYPERTENSIVES CONTROLLED | RECEIVE TREATMENT | YEARLY COST OF OSA | RURAL CHC SPECIALISTS |

*Sources: NFHS-5 hypertension cascade (medRxiv); ResMed economic model, ISPOR Europe 2025; MoHFW Health Dynamics of India 2022-23.*

**Silent signs**
Snoring, fatigue, palpitations dismissed as normal

→

**No routine check**
BP checked rarely; nobody measures at night

→

**Facility-bound**
Cuff at PHC, ECG/Holter, overnight sleep lab

→

**Late detection**
First sign is often a stroke or heart event

### Why spot checks fail

Paroxysmal AF and drops in night-time oxygen happen when nothing is measuring. Sleep studies are expensive and rare outside cities, so 2 in 3 people with hypertension and over 85% of people with sleep apnea are never diagnosed.

**Bottleneck:** every step depends on the patient going to a clinic, so problems are found late.

# User Journey: After State

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| ~8 h | 0 | < 60 s | ≈ ₹100 |
|---|---|---|---|
| DATA PER NIGHT | CLINIC VISITS TO SCREEN | SYNC TO RISK REPORT | HARDWARE PER SCREENING |

**Wear the ring**
No cuff, no clinic, no straps

→

**Passive capture**
Pulse wave, HRV, $SpO_2$, skin temp

→

**Risk flags**
Stiffness, AF-type rhythm, apnea

→

**Early referral**
Report for PHC doctor or teleconsult

## Quantified Efficiency (design targets)

| Metric | Today | NadiSense |
|---|---|---|
| Nights monitored / month | 0–1 | Every night worn |
| Visits needed to screen | 1–3 | 0 (1 to confirm) |
| Time to risk report | Days–weeks | < 60 s after sync |

## Affordable Deployment Model

A PHC or ASHA worker lends one ring out for a week at a time. One ₹4,370 ring can screen about 45 people a year, which is **about ₹100 of hardware per person**. No sleep lab is needed.

**Scope:** NadiSense flags risk and refers to a doctor; it does not diagnose. We will test these targets in Stage 3.

# System Architecture: Block Diagram

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**Sensing**
MAX86141 with red/IR LEDs and a photodiode, TMP117, LIS2DW12

→

**Control & Power**
nRF52840, 12 mAh Li-Po and BQ25120A charger

→

**Firmware Logic**
Continuous at night, short checks by day; skips windows with motion

→

**App & Analytics**
Syncs over BLE, runs the risk models, makes a doctor's report

## Mapping Nadi Pariksha to PPG Features

| Traditional parameter | NadiSense measurement | Screening use |
|---|---|---|
| Vega (rate) | Heart rate from beat detection | Baseline deviation |
| Sthiratva / Tala (stability, rhythm) | Beat-to-beat interval variability | AF-type irregular rhythm |
| Kathinya (arterial hardness) | PPG stiffness index | Hypertension-risk trend |
| Gati / Bala (movement, force) | Waveform shape, pulse amplitude | Personal baseline |
| Tapamana (temperature) | Skin temperature (TMP117) | Fever / illness flag |

**Basis:** Nadi Tarangini studies found the radial pulse looks like a finger PPG wave and linked Kathinya to the stiffness index.

# System Architecture: Target Specs & Hardware

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| ≈ 3 g | 40 µA | 6–10 days | 4 signals |
|---|---|---|---|
| RING WEIGHT | AVERAGE CURRENT | BATTERY PER CHARGE | PPG, $SpO_2$, TEMP, MOTION |

## Engineering Target Specifications & Subsystems

| Parameter | Target Spec | Selected Hardware / Protocol |
|---|---|---|
| Optical sensing | 25 sps, 19-bit ADC, AFE < 25 µA | MAX86141 (SPI) + red 660 / IR 880 nm LEDs + Si PD |
| Heart rate | ±3 bpm at rest vs reference | On-device beat detection, nRF52840 FPU |
| $SpO_2$ | ±3% (70–100%) | Red/IR ratio, calibrated vs reference oximeter |
| Skin temperature | ±0.1 °C sensor accuracy | TMP117, I²C, one reading per minute |
| Motion rejection | Gate windows above motion limit | LIS2DW12 accelerometer (~1 µA) |
| Power & battery | 40 µA average, 6–10 days per charge | 12 mAh curved Li-Po (245 mm³), BQ25120A |
| Wireless | One batched sync per night | BLE 5.0, custom GATT service |
| Size & weight | ID 19 mm, OD 26 mm, 8 mm wide; ≈ 3 g | SLA resin shell, polyimide flex PCB |

**Why MAX86141:** it draws 11 µA where the MAX30101 drew 600 µA, and it needs no 5 V rail for a green LED.

# CAD Design & Mechanical Specs

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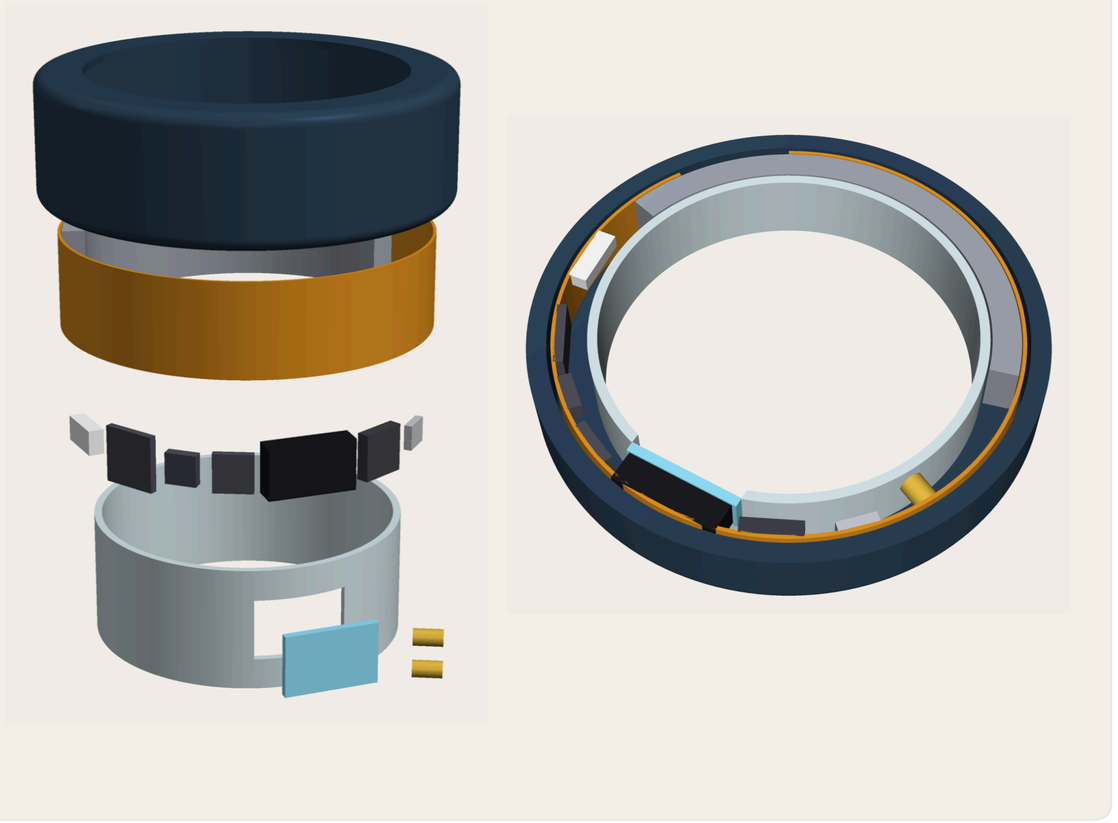


*Figure: exploded assembly and half-section, ring v2 (AI Generated)*

## Mechanical Design & Material Selection

- **Envelope:** ID 19 mm, OD 26 mm, 8 mm wide; wall = 0.5 liner + 2.0 cavity + 1.0 shell.
- **Shell:** biocompatible SLA resin U-channel, 0.8 mm outer and 0.3 mm inner fillets.
- **Liner:** clear resin with a 6 × 4 mm optical window sealed in clear epoxy.
- **Layout:** 245 mm³ curved Li-Po on the back of the finger; optics on the palm side; flex PCB wraps 330°.
- **Mass:** ≈ 3 g estimated from CAD volumes.

**Design Iteration:** we first planned a 20–40 mAh cell, but the CAD cavity holds only 245 mm³, so we resized it to 12 mAh.

# CAD Design: FEA Stress Analysis

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| 5.9 | 8.5 MPa | 0.12 mm | 0.8% |
|---|---|---|---|
| FACTOR OF SAFETY | PEAK VON MISES | MAX DEFLECTION | MESH CONVERGENCE |

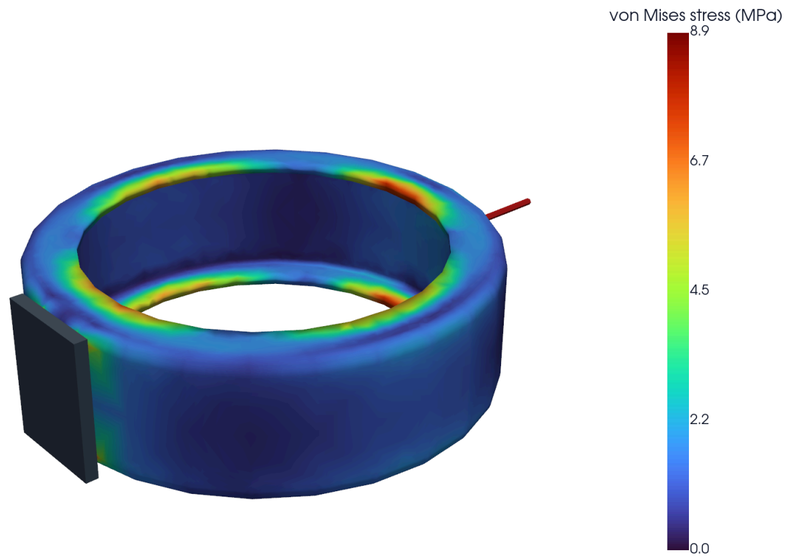



*Figure: von Mises stress, 50 N squeeze, 90k-element mesh (AI Generated)*

## Mechanical Validation & FEA Report

- **Load:** 50 N squeeze across the ring; 6 mm support patch opposite.
- **Material:** rigid SLA resin, E = 2.5 GPa, UTS 50 MPa (assumed); shell only, conservative.

| Mesh | Elements | DOFs | Peak σ |
|---|---|---|---|
| Coarse | 22,945 | 135,009 | 9.35 MPa |
| Medium | 61,197 | 329,184 | 8.53 MPa |
| Fine | 90,507 | 457,659 | 8.46 MPa |

**Design Iteration:** we added 0.3 mm fillets at the inner channel corners before the analysis to remove sharp stress points.

# Electronics Schematics & Power Budget

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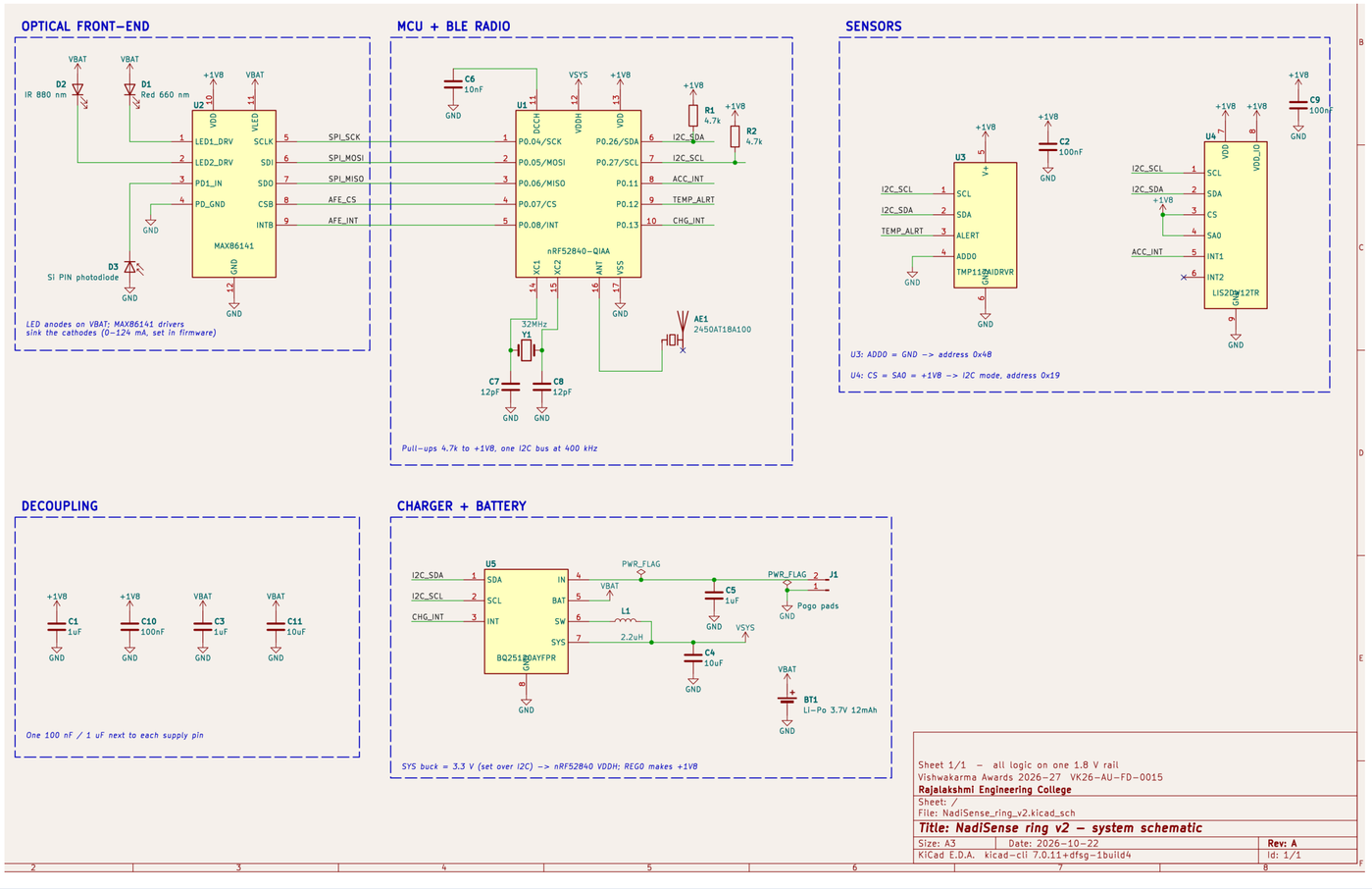



*Figure: KiCad 7 schematic sheet, Rev A, ERC clean (AI Generated)*

## Power Budget

| Mode | Avg current |
|---|---|
| Night, 8 h (continuous) | 98 µA |
| Day, 16 h (spot checks) | 12 µA |
| **24 h average** | **40 µA** |
| Life, 12 mAh × 80% | 9.9 days |
| Worst case* | 6.0 days |

### Night current breakdown (98 µA)



■ LEDs 60 µA ■ AFE 22 µA
■ MCU 10 µA ■ Other 7 µA

*20 mA LEDs, 2× CPU time

**Power Distribution:** the BQ25120A buck gives 3.3 V to the nRF52840, whose regulator makes 1.8 V for all logic. LEDs run off the battery.

# Electronics BOM & Procurement

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## Itemized Bill of Materials (one ring)

| Ref | Component | Part / Specification | Sourcing / Lead Time | Est. (INR) |
|---|---|---|---|---|
| U1 | BLE SoC | nRF52840, WLCSP | Distributor (5–7 days) | ₹420 |
| U2 | Optical AFE | MAX86141, WLP | Distributor (5–7 days) | ₹520 |
| U3 | Temperature | TMP117, ±0.1 °C | Distributor (5–7 days) | ₹260 |
| U4 | Accelerometer | LIS2DW12 | Distributor (5–7 days) | ₹140 |
| U5 | Charger | BQ25120A | Distributor (5–7 days) | ₹180 |
| D1–D3 | Optics | Red/IR SMD LEDs, Si photodiode | Distributor (5–7 days) | ₹220 |
| Y1, AE1, L1 | RF + passives | 32 MHz crystal, chip antenna, 2.2 µH, R/C | Distributor (5–7 days) | ₹230 |
| PCB | Flex PCB | 2-layer polyimide, 0.2 mm | Imported (10–14 days) | ₹700 |
| BT1 | Battery | 12 mAh curved Li-Po (custom) | Imported (3–4 weeks) | ₹1,200 |
| MECH | Shell, liner, potting | SLA resin, epoxy, silicone, pogo pads | Local (4 days) | ₹500 |
| | **Total per unit** | | | **₹4,370** |

**Procurement Risk:** the custom curved cell has the longest lead time, so it is ordered first, in Month 1.

# Your Maker Journey

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*Figure: Team Ring of Care, REC (AI Generated)*

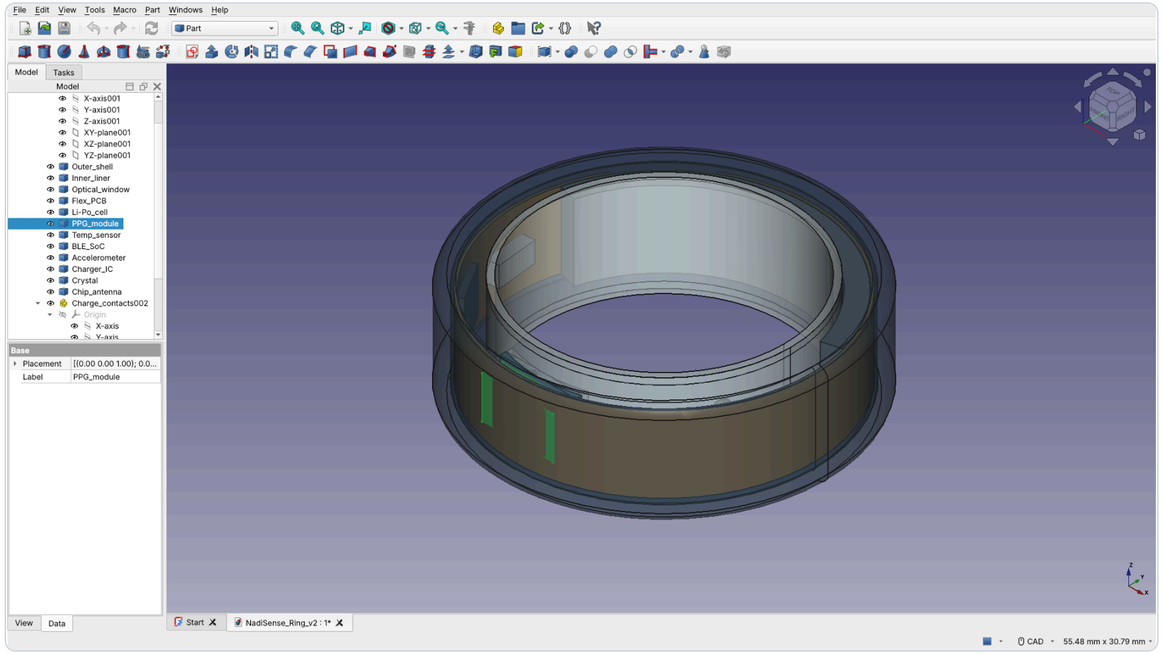
## Stage 1 to Stage 2 Design Iterations

- **Form factor:** moved from a handheld reader to a ring so we can record through the night.
- **Sensing:** added $SpO_2$, skin temperature and motion sensing to the PPG.
- **Optical AFE:** the power budget ruled out the MAX30101 (600 µA), so we moved to the MAX86141 (11 µA).
- **Battery:** cut from 20–40 mAh to 12 mAh to fit the CAD cavity; still 6–10 days per charge.
- **Shell:** 0.3 mm internal fillets added; FEA safety factor 5.9 at 50 N.
- **Software:** companion app live at nadisense-app.vercel.app in 8 Indian languages, running on simulated sensor data until the ring is built.
- **Repository:** github.com/yeswantsakthi-hub/NadiSense, with STEP files, KiCad project, FEA scripts, power budget and app code.

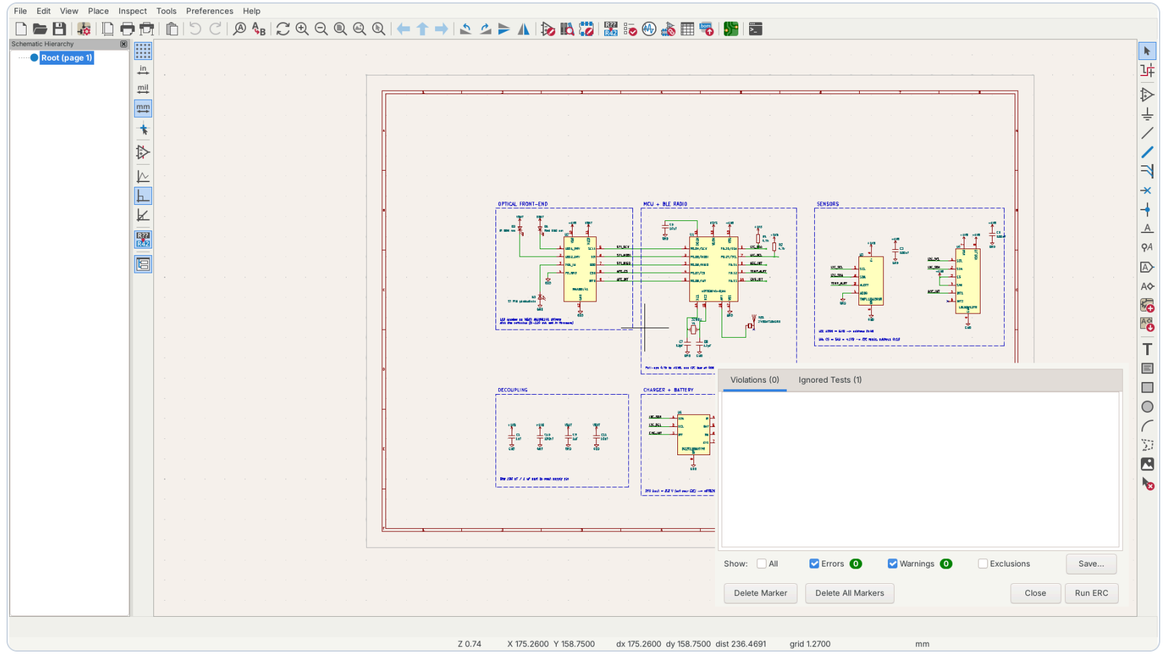
**Institutional Sign-Off:** design reviewed by Faculty Guide [Name], Dept. of [ ], Rajalakshmi Engineering College, Chennai.

# Your Maker Journey: Design Tools

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*FreeCAD 1.0: ring v2 STEP assembly, 13 parts in the model tree (AI Generated)*




*KiCad 7: Electrical Rules Check, 0 errors, 0 warnings (AI Generated)*

**CAD:** parametric model exported to STEP and checked in FreeCAD; same file used for the FEA.

**ERC:** first run found 6 errors and 14 warnings; we added power flags, fixed pin types and a project library.

**Files:** STEP, KiCad project, ERC report and netlist are in github.com/yeswantsakthi-hub/NadiSense.

# Safety & Hazard Mitigation

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## Design FMEA (Severity × Occurrence × Detection, each 1–10)

| Failure Mode | Potential Hazard | Embedded Fail-Safe / Mitigation | S | O | D | RPN |
|---|---|---|---|---|---|---|
| Cell overheat | Burn at the finger | Charger thermal limit; cut-off above 43 °C | 9 | 2 | 3 | 54 |
| LED skin heating | Warmth, discomfort | LED current cap; duty-cycled sampling | 5 | 2 | 4 | 40 |
| Motion artifact | False alert | IMU gating; alert only on repeat episodes | 4 | 6 | 4 | 96 |
| Missed event | False reassurance | Never shows “all clear”; clinical check | 8 | 4 | 6 | **192** |
| Data exposure | Privacy breach | On-phone processing; encrypted BLE and storage | 6 | 2 | 5 | 60 |

### Physical Fail-safes

Fully potted electronics, rounded edges, no exposed conductors except gold charging pads; ship mode on the charger.

### Standards Compliance

IEC 60601-1-11, ISO 80601-2-61, ISO 10993, IEC 62133-2, IEC 62304, CDSCO MDR 2017, DPDP Act 2023

**Highest RPN:** missed event (192). The app never says “all clear”, and every flag ends in a referral.

# Roadmap: 4-Month Fabrication Plan

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| | Nov 2026 | Dec 2026 | Jan 2027 | Feb 2027 |
|---|---|---|---|---|
| **Procurement** | Custom Li-Po, ICs, flex PCB order | | | |
| **Mechanical** | Print shells, fit trials, drop test | | | |
| **Electronics** | MAX86141 eval kit, flex PCB layout, ring v2 build | | | |
| **Firmware & App** | | Algorithms on public PPG data, BLE sync, reports | | |
| **Validation** | | | Bench tests, then overnight pilot | |
| **Milestone** | | | | **Stage 3 demo ready** |

**Stage 3 Target:** a working ring streaming live HR, $SpO_2$ and temperature, producing a screening report from one night's wear.

# Technical Challenges & Risks

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### Motion Artifacts

**Risk:** finger movement corrupts the PPG signal far more than wrist placement; daytime readings may be unusable.

### $SpO_2$ Skin-Tone Bias

**Risk:** pulse oximeters are documented to overestimate $SpO_2$ in darker-skinned patients, which matters for Indian users.

### Power vs Sensing

**Risk:** the cavity fits only a 12 mAh cell; the first AFE choice (600 µA) would drain it in under a day.

### No Labelled Nadi Data

**Risk:** no public dataset links PPG to traditional pulse readings, and PPG alone cannot measure absolute blood pressure.

**Verification Plan:** every risk is tested on the bench against a reference oximeter and BP cuff in January 2027.

# Potential Solutions & Risk Mitigation

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### Motion Artifacts

**Mitigation:** IMU-gated windows; screen mainly during sleep, when the hand is still.

**Fallback:** IR-only HR in daytime; tighter fit with a sizing kit.

### $SpO_2$ Skin-Tone Bias

**Mitigation:** calibrate against a reference oximeter across varied skin tones.

**Fallback:** flag apnea from relative desaturation drops, not absolute $SpO_2$.

### Battery Life

**Mitigation:** MAX86141 at 25 sps and one BLE sync per night; 6–10 days by our power budget.

**Fallback:** night-only mode; 2.5 mm cavity for a ~16 mAh cell.

### Data Gap

**Mitigation:** train on public datasets (PPG-DaLiA, PPG-BP, MIMIC waveforms, MESA).

**Fallback:** rule-based HRV and desaturation scores; collect our own labelled pilot data.

**Scope:** NadiSense tracks arterial-stiffness *trends* and refers for a cuff reading; it never reports a BP number.